

Reciprocal AI-Neuroscience Translation:

From Biological Mechanism to Engineered Failure, Rescue, and Biological Retest

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Abstract

Neuroscience has repeatedly inspired artificial intelligence, while artificial models increasingly predict neural responses, generate stimuli, and participate in closed-loop experiments. These achievements do not by themselves establish that an engineered system preserves the biological mechanism that motivated it. A model can fit intact observations yet fail under lesion, drift, compensation, or rescue; a restored mean can conceal a relation-wrong rescue; and an artificial result cannot authorize a biological intervention unless the map back to the preparation is explicit. I propose **reciprocal AI-neuroscience translation** as a typed experimental protocol for closing this gap. A biological relation is described by its sender, receiver, state, delay, capacity, update rule, scale, and consequence. A partial translation map declares exactly which relations an engineered analogue is intended to preserve. Capacity-matched alternatives are then compared on held-out perturbations, compensation, correct rescue, wrong rescue, off-target preservation, and mapping drift. Only the resulting discriminating signature is returned to a matching-scale biological experiment, after which both models must be revised.

The proposal extends a documented Self-Aware Networks (SAN) genealogy from an embodied “learning loop that learns itself” in 2011, through same-network production and reception in 2012, neural read/write protocols in 2017, a cortical-traffic return question in 2021, and a mature reciprocal method in 2026. It also credits established precedents: dendritic computation, inception-loop experiments, digital twins, functional connectomics, adaptive stimulation, neuroprosthetic coadaptation, sensory substitution, biohybrid systems, and adversarial AI for mechanistic hypothesis generation. The residual contribution is not “AI inspired by biology.” It is the seven-operation acceptance protocol joining typed abstraction, held-out engineered failure, selective rescue, and a predeclared biological return test.

A deterministic reference application evaluates 36 paired synthetic transition systems, seven intervention families, and four models, yielding 1,008 model-intervention evaluations. The correctly typed model achieved mean intervention-effect root-mean-square error (RMSE) 0.01454, compared with 0.37497 for both endpoint baselines and 0.46317 for a deliberately wrong typed map. Correct rescue restored the declared synthetic target exactly by construction, whereas every mean-matched wrong rescue retained conditional-signature RMSE of at least 0.13352. Fifteen validators and a byte-identical rerun passed. Lean 4.32.0 also machine-checked eight bounded logical leaves with no admitted proofs. These results establish properties of the declared formal and synthetic models, not biological validation. The defining empirical gate remains a successful, safe, matching-scale return experiment that distinguishes the proposed mechanism from its strongest alternatives.

Keywords: AI for neuroscience; mechanistic translation; digital twin; closed loop; intervention; rescue; neural drift; sensory substitution; biohybrid system; Self-Aware Networks

1 Introduction

Biology has served as both inspiration and test bed for computation. McCulloch and Pitts abstracted neural activity into a logical calculus, and Rosenblatt made weighted units learnable [1,2]. Later artificial networks departed far from literal neuronal anatomy while gaining extraordinary practical power. In the opposite direction, contemporary models can predict neural responses, synthesize inputs expected to drive particular neurons, guide closed-loop stimulation, and expose hypotheses that can be compared with human and animal evidence [8-12,18]. This traffic is productive precisely because an abstraction need not duplicate its source.

The freedom to abstract creates a validation problem. If an engineered model reproduces an observed endpoint, what has actually transferred? It may have preserved a causal relation, approximated a sufficient statistic, exploited a correlational shortcut, or reached the same endpoint by a different route. Those possibilities can remain indistinguishable on intact observations. They often separate only when a relation is removed, delayed, remapped, compensated for, or selectively restored.

The same problem appears on the return trip. A successful artificial lesion or rescue does not automatically specify a safe biological intervention. The engineered variable may have no unique biological counterpart; a map that once worked may drift; a receiver may learn a new contingency; or a rescue may restore one summary while damaging unmeasured structure. The return therefore needs a declared preparation, scale, perturbation, readout, correct rescue, wrong rescue, off-target control, stopping rule, and revision rule.

I propose reciprocal AI-neuroscience translation as a protocol in which the **round trip**, rather than the initial analogy, is the unit of evidence:

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biological relation
-> typed partial abstraction
-> capacity-matched engineered analogue
-> held-out failure, drift, compensation, and rescue
-> discriminating signature
-> matching-scale biological retest
-> revision of both models

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The proposal makes three commitments. First, translation maps are partial and falsifiable; they never silently establish substrate identity. Second, engineered success must survive interventions that distinguish mechanisms, including a relation-wrong rescue that matches a coarse endpoint. Third, the full protocol is incomplete until the discriminating result is returned to biology. The application and formal proofs reported here validate tools for designing that process. They do not claim to complete the biological return.

2 The Established Landscape and the Residual Problem

2.1 Richer biological sources

Early neural abstractions deliberately discarded most cellular detail. Modern neuroscience shows why a new translation protocol cannot treat a neuron as an untyped scalar node. Dendrites integrate inputs nonlinearly and in a location-dependent way [3]. Cortical dendrites can discriminate input sequences in bounded preparations [4]. Segregated dendritic compartments can support biologically motivated credit-assignment mechanisms in engineered models [5]. A translation may legitimately ignore these relations for

one question, but it must say so; otherwise a missing mechanism can be mistaken for an experimentally supported equivalence.

The relevant unit is therefore a receiver-relative relation, not a visual resemblance between diagrams. A connection has a sender, a receiver, a state, a delay, a capacity, an update rule, and a consequence. Morphology, compartment, history, inhibition, and plasticity can change what the same arriving signal does. This does not require every engineered model to reproduce every biological variable. It requires the omitted variables and preserved relations to be named.

2.2 Model-to-neuron return already exists

Walker and colleagues’ “inception loop” is a decisive precedent [8]. A predictive model of mouse primary visual cortex generated most-exciting inputs, and the stimuli were presented back to the same neurons *in vivo*. Later work on functional bipartite invariance used a digital twin to synthesize varied exciting inputs and test them biologically [9]. MICrONS-linked work related validated digital-twin tuning to measured connectivity while retaining the need for independent validation in new stimulus domains [10]. These studies demonstrate that an engineered model can generate a biological return experiment.

The residual is not the idea of returning a model-generated stimulus to biology. It is a general acceptance discipline for deciding which relation earned that return, which alternative mechanisms remain, how correct and wrong rescue are separated, and how a negative result revises both models. A model-predicted maximally exciting stimulus can be valuable even when the model is not a complete mechanistic copy. Reciprocal translation asks a narrower additional question: which perturbation signature would distinguish the proposed relation from another system with similar intact predictions?

2.3 Adaptive, coadaptive, and biohybrid loops

Real-time experimental systems already select stimuli and stimulation from incoming data. The improv platform demonstrates online prediction, Bayesian stimulus selection, and closed-loop optogenetic photostimulation in zebrafish [11]. Neuroprosthetic work shows that decoder adaptation and biological plasticity can jointly shape skilled control [12]. Sensory-substitution studies show that a learned contingency can make information from an unusual interface behaviorally usable [13-15]. BioemuS supplies a configurable real-time biomimetic spiking component for biohybrid experiments [16].

These results make drift and coadaptation central rather than optional. A stable endpoint after learning can arise because the engineered decoder changed, because the organism changed, or because both changed. The endpoint cannot identify which relation was restored. A protocol must measure mapping change, recovery time, conditional response structure, and off-target effects, and it must distinguish restoration of the nominated relation from compensation through a new route.

2.4 AI-generated hypotheses

Toker and colleagues recently used adversarial AI to generate mechanistic and treatment hypotheses for disorders of consciousness and compared different claims against reported human and animal evidence [17]. This is a strong current example of AI participating in scientific mechanism generation. It also illustrates the importance of keeping evidence classes separate: model output, retrospective human evidence, animal evidence, and causal intervention do not become interchangeable because they appear in one study.

Anthropic’s feature-amplification experiments supply a useful engineering comparator [18]. Artificial feature intervention can produce broad semantic spillover, showing how a local control hypothesis can have global and unexpected consequences. It is not neuroscience evidence. It is a reminder that a successful targeted intervention needs off-target measurement and a wrong-rescue control.

2.5 Residual contribution

The literature therefore contains most component operations. The paper’s residual is the following joined acceptance protocol:

1. type the preparation-specific biological relation;
2. declare a partial translation map and its exclusions;
3. construct a capacity-matched engineered analogue and strong alternatives;
4. test unseen lesions, delay changes, drift, and compensation;
5. compare correct rescue with a type-compatible but relation-wrong or mean-matched rescue;
6. translate only the discriminating signature back to the corresponding biological scale; and
7. revise or retire both models under predeclared criteria.

If a prior paper is found that already specifies and evaluates this complete seven-operation method, the novelty claim should be narrowed. The individual operations are not claimed as new.

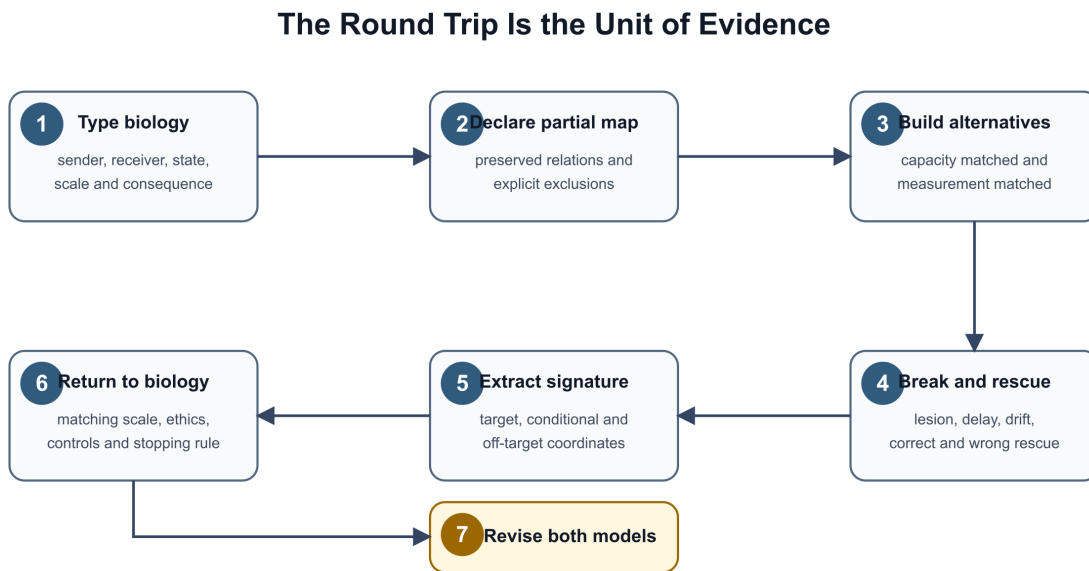


Figure 1. The complete round trip is the claimed unit of method. Stages 1-6 may succeed or fail separately; Stage 7 requires both models to change when the return result disagrees with the proposed correspondence.

3 Source-Faithful SAN Genealogy

The mature protocol is a 2026 synthesis. Its earlier sources establish narrower ancestors and are not backdated.

In a September 21, 2011 source, I described an embodied cycle in which sensation changes a statistical

learning and prediction system, neural activity contributes to action, and returned sensation is associated back with the continuing process. The phrase “a learning loop that learns itself” names the whole active loop rather than a detached self-symbol. In April 2012 I added a same-substrate sender/receiver intuition: the cells producing the ongoing learned “song” also listen to it. These sources establish recurrent production, reception, action, and learning. They do not contain digital twins, formal translation maps, PWD, NAPOT, or the present acceptance protocol.

The 2017 *Neural Lace* discussions contrasted connection-and-feedback with connectionless communication and proposed an Alternating Transmission Protocol that could switch sending, receiving, connection, and feedback relations. A March 25, 2021 dialogue then traced reverse, vertical, horizontal, cortical, and thalamic traffic while asking whether a repeatable sensory trajectory could be decoded and a compatible pattern returned as artificial touch. The 2022 SAN outline placed Neo Mind Cycle/BCI work, biological Self Aware Networks, and artificial neural systems inside one program. None of those documents reports a completed biological-write experiment.

Book 1 in 2024 joined molecular mechanisms, cells, oscillations, memory, rendering, action, observation, and consciousness. *Building Sentient Beings* in 2025 developed the engineering route through active internal state, embodiment, development, and self-modeling [19]. Cycle 37 in 2026 made the reciprocal scientific method explicit: recover a living operation, state its engineering requirement, construct it, test controlled loss and rescue, and return the discriminating result as a sharper biological experiment. That last step is the mature contribution evaluated here.

This chronology does two things. It preserves the long-running biological and engineering question without pretending the final formalism existed in the early notes. It also locates authorship correctly: external neuroscience establishes the measured components; SAN contributes the documented joined method and the present formal, synthetic, and experimental program.

4 Typed Reciprocal Translation

4.1 Biological and engineered systems

Let the biological and engineered systems be finite typed transition systems

$$B = (V_B, E_B, X_B, U_B, Y_B, F_B), \quad E = (V_E, E_E, X_E, U_E, Y_E, F_E). \quad (1)$$

The node and edge sets do not carry meaning by position alone. Each relation is annotated with a type tuple

$$\tau(e) = (\text{sender}, \text{receiver}, \text{state}, \text{delay}, \text{capacity}, \text{update}, \text{scale}, \text{consequence}). \quad (2)$$

A biological relation may be molecular, synaptic, dendritic, cellular, laminar, population-level, behavioral, or embodied. The engineered partner may use a different substrate and state representation. Comparison is licensed only for the fields declared commensurable.

4.2 Partial translation maps

A translation map

$$\phi : B \rightarrow E \quad (3)$$

is partial. Its domain names the biological states and relations being abstracted; its codomain names their engineered representation. It should include a typed measurement map rather than relying on vocabulary alone. If a source delay is mapped to an engineered gain, for example, the correspondence is ill typed unless an explicit experiment establishes that the gain is the declared proxy for the delay-dependent consequence.

The return map $\psi : E \rightarrow B$ is separately declared. Neither map is assumed invertible. A left inverse on one target observable is weaker than an isomorphism of the systems. This matters because biological state includes unmeasured history, metabolism, development, and plasticity that an engineered analogue may not represent.

4.3 Intervention signatures

For an intervention i , let a declared measurement operator M produce the effect signature

$$\Delta_i = M(F^{(i)}(X)) - M(F^{(0)}(X)). \quad (4)$$

The signature should include more than the endpoint used for training. Depending on the preparation, it may include conditional slopes, latency, variance, receiver selectivity, recovery dynamics, off-target effects, and structural or state variables. Biological and engineered signatures are compared only after explicit conversion to dimensionless or otherwise commensurable coordinates:

$$D_i = d(\psi_B(\Delta_i^B), \psi_E(\Delta_i^E)). \quad (5)$$

A small D_i supports the named signature correspondence. It does not prove that the substrates, latent states, or organism-level functions are identical.

4.4 Selective rescue

A rescue must restore the target while preserving declared off-target measurements:

$$R_i = \|M_T(F^{(i,r)}) - M_T(F^{(0)})\|, \quad O_i = \|M_{-T}(F^{(i,r)}) - M_{-T}(F^{(0)})\|. \quad (6)$$

A selective rescue requires $R_i \leq \epsilon_T$, $O_i \leq \epsilon_O$, and rejection of a predeclared wrong rescue. Restoring a mean is not enough when multiple relations share that mean. A correct rescue is therefore defined by a discriminating signature, not by semantic confidence.

4.5 Return authorization

An engineered result earns a biological return test only when the following fields are frozen: preparation; biological scale; mapped variable; perturbation; correct rescue; relation-wrong rescue; readouts; off-target control; safety constraint; expected signatures under the strongest alternatives; stopping rule; and revision rule. This is a research-authorization standard, not a clinical authorization standard. Any human or animal intervention remains subject to the applicable ethics, safety, and regulatory review.

5 Formal Results and Countermodels

5.1 Observational fit does not determine intervention response

There exist two functions $f, g : \{0, 1\} \rightarrow \mathbb{N}$ for which

$$f(0) = g(0), \quad f(1) \neq g(1). \quad (7)$$

Agreement on the observed state therefore does not entail agreement on an unobserved intervention. This finite witness is elementary, but it fixes the inference boundary: observational fit alone cannot authorize a unique causal recommendation.

5.2 Conditional mapped rescue

Suppose $\phi : B \rightarrow E$ and $\psi : E \rightarrow B$ satisfy

$$\psi(\phi(b)) = b \quad (8)$$

on the declared target domain. If an engineered rescue restores $\phi(b^*)$, decoding restores b^* . The result is conditional on the left-inverse premise. It neither identifies nor empirically validates that premise.

5.3 Wrong-rescue discrimination

If a signature S is injective on the candidate rescue set, then

$$w \neq b^* \implies S(w) \neq S(b^*). \quad (9)$$

This result explains why a rescue signature must contain coordinates that can distinguish wrong relations. A non-injective endpoint, such as one mean, permits false rescue.

5.4 Round-trip ceiling

Even a perfect encode-decode round trip preserves only the observable named in the premise. It does not establish equality of unmeasured state, learning history, morphology, metabolism, consciousness, or organism-level function.

5.5 Machine-check receipt

Lean 4.32.0 with Mathlib v4.32.0 compiled eight bounded theorem leaves with exit code 0. A whole-word scan found no sorry or admit. The type-compatibility leaves use Lean's standard quotient, choice, and propositional-extensionality infrastructure; the finite countermodels require propositional extensionality; the mapped rescue, wrong-rescue, off-target, and round-trip leaves report no axioms. The exact command, source hash, and theorem-by-theorem audit are preserved in `formal/FORMAL-CHECK-RECEIPT-20260825.md`.

The formal result is deliberately narrower than the paper. It proves conditional logical properties of declared maps and signatures. It does not identify a biological map, validate a generator as neuroscience, or establish that a proposed rescue is effective or safe.

6 Reference Application

6.1 Frozen design

The executable pilot generates paired deterministic “biological-like” and engineered systems with the chain

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source/context -> learned state
-> routed state with delay
-> typed receiver consequence
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Thirty-six paired systems are evaluated under seven held-out intervention families: route lesion, state-path lesion, delay reversal, receiver drift, correct route rescue, mean-matched wrong rescue, and combined lesion. The engineered representation preserves declared internal relations under nonzero scale maps. Training uses intact observational trajectories only.

Four models are compared:

1. a correlational ridge endpoint baseline;
2. a 30-parameter polynomial endpoint black box;
3. a 15-parameter typed model with measured intermediate states and paired measurements for map estimation; and
4. a 15-parameter wrong-map typed control in which state and route roles are deliberately exchanged.

The endpoint baselines receive source, context, receiver identity, and endpoint data but no intervention variable or internal graph typing. This asymmetry is intentional and disclosed: the experiment asks whether typed internal measurements make unseen interventions identifiable in a generator built from those relations. It does not show that all black boxes lack such capacity. A black box trained on sufficiently varied interventions or internal measurements could succeed.

6.2 Metrics and predeclared gates

The primary metrics are held-out intervention-effect RMSE, endpoint RMSE, effect-calibration slope, translation-map error, correct-rescue recovery, wrong-rescue conditional-signature error, and determinism. The pilot passes only if the correctly typed model improves held-out intervention and rescue prediction, the wrong typed map loses that advantage, correct rescue restores the target, wrong rescue fails a conditional signature, and an independent rerun reproduces the frozen artifacts.

6.3 Reproducibility

The configuration, generator, analysis, validation code, complete 1,008-row table, diagnostic table, figures, run manifest, and deterministic rerun are under application-proof/. The reported and rerun outputs are byte-identical. Fifteen validators check configuration and code hashes, row and model completeness,

effect performance, the wrong-map control, rescue behavior, map recovery, provenance, finite values, and determinism.

7 Results

7.1 Held-out intervention effects

Across the six non-rescue held-out intervention families used in the aggregate error summary (n=216 system-intervention cells per model), mean engineered intervention-effect RMSE was:

Model	Parameters	Mean effect RMSE
Correlational endpoint model	10	0.374974
Polynomial endpoint black box	30	0.374974
Correct typed relation model	15	0.014542
Wrong-map typed control	15	0.463173

The typed-to-best-endpoint error ratio was 0.03878. The additional polynomial capacity did not help because the intact training inputs did not identify how the unobserved intervention acted. The result is therefore a demonstration of information and inductive-bias value in this frozen family, not a universal ranking of interpretable and black-box models.

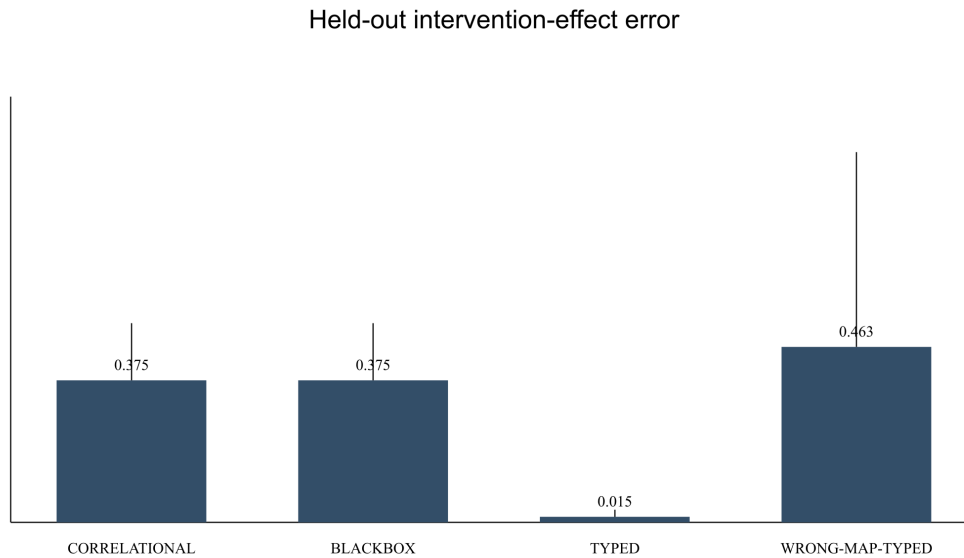


Figure 2. Mean held-out intervention-effect RMSE. The correctly typed model receives the internal variables needed by the synthetic mechanism; the wrong-map ablation tests whether merely possessing intermediate variables is sufficient. Bars summarize synthetic system-intervention cells descriptively; they are not inferential statistics for tissue, animals, or people.

7.2 Endpoint error and calibration

The correctly typed model retained its advantage at the engineered endpoint. Its mean effect-calibration slope was 0.985. The endpoint-only models returned zero predicted intervention effect because no intervention structure appeared in their training input. The recovered translation-map mean absolute error was 0.000623.

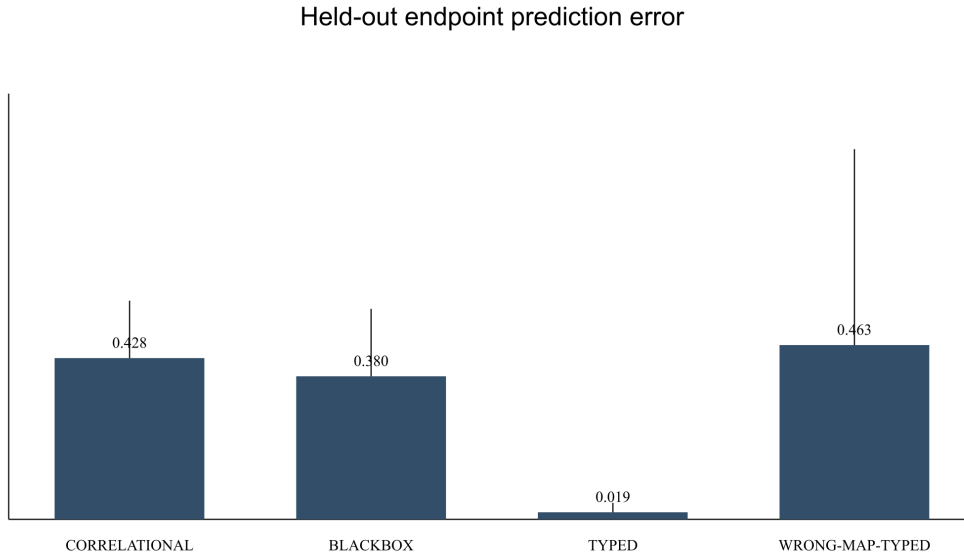


Figure 3. Endpoint RMSE across the frozen intervention set. The figure should be interpreted with the design asymmetry stated in Section 6.1. Means and standard deviations are descriptive summaries of the frozen synthetic systems.

7.3 Correct and wrong rescue

The correct route rescue restored the intact synthetic target with maximum RMSE 0 by construction. The more informative control was the mean-matched wrong rescue: it restored a coarse mean while preserving the wrong conditional relation. Every such trial retained a conditional-signature RMSE of at least 0.133522. This is the central application result. A recovered endpoint can coexist with a failed mechanism signature.

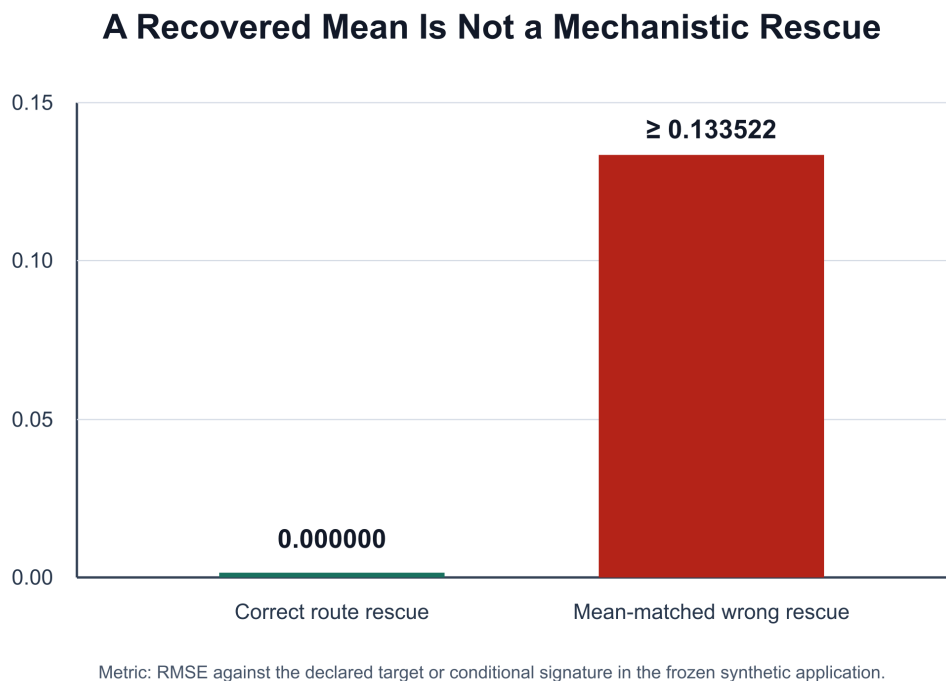


Figure 4. The exact correct-rescue result is built into the synthetic intervention; the wrong-rescue result tests whether a recovered mean can pass the richer conditional signature. Values are application measurements only.

7.4 Validation outcome

All 15 validators passed, and the deterministic rerun reproduced every declared result artifact byte-for-byte. No missing model, intervention, nonfinite value, provenance mismatch, or seed drift was detected. Passing validation establishes internal reproducibility of the frozen synthetic pilot; it is not an external replication.

8 What the Pilot Changes-and What It Does Not

The pilot supports three bounded methodological claims. First, observational performance is an insufficient acceptance gate for causal translation when unobserved interventions are not identified by the training inputs. Second, correctly typed internal measurements can enable extrapolation to lesions and rescues in a system generated by those relations. Third, rescue should be evaluated with conditional and off-target signatures rather than one recovered summary.

The pilot does not show that its four-stage graph is biologically complete. It does not show that black-box models cannot infer interventions when trained on appropriate interventional data. It does not compare every causal-discovery, state-space, or system-identification method. It does not demonstrate safe stimulation, conscious-state control, clinical utility, or a biological return. The generator was intentionally designed around the nominated relation, so the typed model's advantage is evidence that the test can detect correct and incorrect maps in this family-not evidence that the same map exists in a brain.

The wrong-map control is particularly important. If both typed models had succeeded, the “typed” advantage

could have reflected extra features rather than correct relational semantics. Its failure localizes the advantage to the role assignment within the generator. A future benchmark should add capacity-matched causal state-space models, train alternatives on partial interventions, vary observability, introduce stochastic hidden confounding, and test whether signature choice remains discriminating.

9 Biological Return Experiments

9.1 Visual inception-loop extension

Preparation: identified neurons in mouse V1 under an established predictive-model protocol.

Engineered divergence: fit two models that agree on intact or most-exciting-input responses but posit different receptive-field relations.

Held-out perturbation: optimize a stimulus family on which the two relations make opposing cell-specific predictions.

Correct-rescue analogue: restore the model-nominated relation in stimulus space.

Wrong rescue: match mean predicted firing while altering the conditional feature relation.

Readout: same-cell response, latency, trial distribution, receptive-field conditional structure, and neighboring-cell off-target activity.

Falsifier: the discriminating family fails to separate the models, or a simpler model predicts the return equally well.

This experiment extends rather than supersedes inception-loop work. Its contribution would be the predeclared relation-wrong rescue and two-model revision rule.

9.2 Drift-aware sensory substitution

Preparation: a learned noninvasive artificial feedback contingency, initially avoiding any claim of arbitrary percept writing.

Models: fixed-address decoder, adaptive endpoint model, and receiver-relative typed map.

Perturbation: controlled remapping or decoder drift introduced after criterion performance.

Rescue: recalibrate the nominated receiver relation while holding overall feedback magnitude constant.

Wrong rescue: restore mean feedback through a mismatched channel or temporal contingency.

Readout: behavior, calibration, remapping time, transfer, aftereffects, and off-target perceptual reports.

Falsifier: the typed map provides no reproducible advantage over a simpler adaptive decoder.

The ethical boundary is strict: sensory substitution supports learned use of alternative signals, not arbitrary insertion of a chosen conscious scene.

9.3 Dendritic or laminar return

Preparation: a preparation in which compartment-specific input, local dendritic state, output, and a causal perturbation can be measured.

Engineered analogue: a compartmental model and capacity-matched rate/recurrent alternatives.

Perturbation: lesion or delay the nominated compartmental route.

Correct rescue: restore the route's state-dependent contribution.

Wrong rescue: match output rate by global gain while leaving the compartmental relation absent.

Readout: local voltage or calcium, output timing, receiver response, and off-target compartments.

Falsifier: the relation-wrong rescue restores the complete signature or the compartmental model fails held-out perturbations.

9.4 Return-test registration

Before data collection, each experiment should register the mapping table, tolerated signature distance, sample and exclusion rules, generalization unit, alternatives, adverse-event boundary, and model-revision rule. A negative return is informative when it localizes the failed map. It should not be redescribed as general support for the larger architecture.

10 System Context and Data Architecture

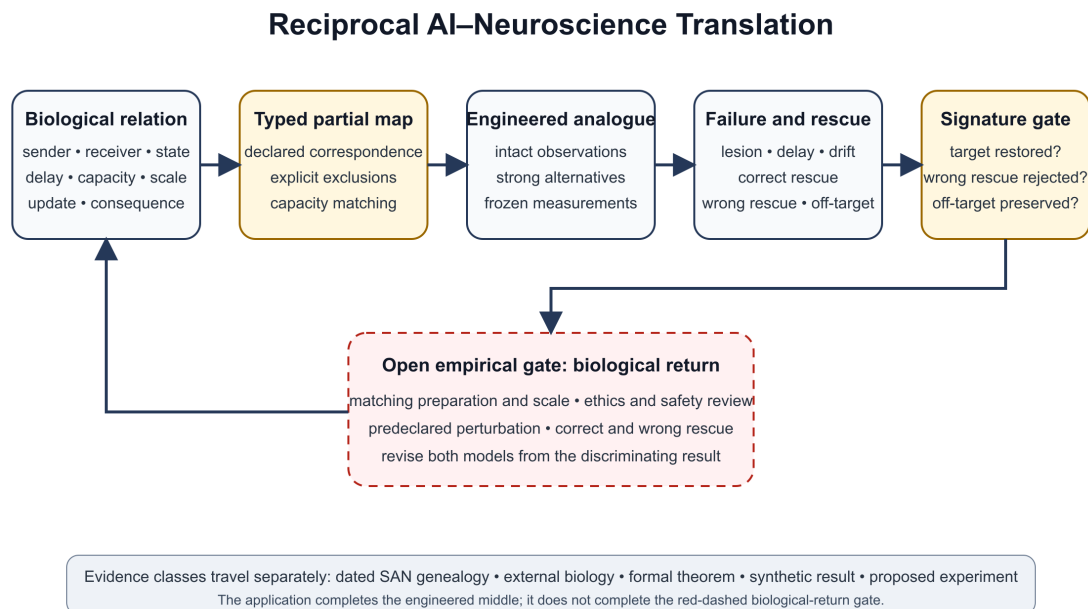


Figure 5. Governed evidence flow. The red-dashed biological-return box remains open because this paper does not report an in-vivo return experiment.

The research system has five governed surfaces: dated biological sources, external primary literature, a typed hypothesis model, an executable engineered test bed, and a biological-return specification. Evidence moves forward only through a declared interface. Provenance moves with every claim so a formal theorem cannot silently become an empirical result and a synthetic metric cannot silently become a biological observation.

The application implements the middle of this architecture. Frozen configuration generates paired systems; intact observations train four candidate models; a withheld intervention engine applies lesion, drift, delay, and rescue families; a signature evaluator measures target and off-target structure; and an independent validator compares the reported run with a deterministic rerun. The unexecuted biological-return module is shown as an open gate, not as a completed result.

11 Strongest Alternatives and Retirement Rules

The strongest black-box alternative is not “a model with no mechanism.” It is a high-capacity predictor trained on broad observational and interventional data, perhaps with latent state sufficient for transfer. Such a system may outperform a human-readable typed model. If it also distinguishes correct and wrong rescue, remains calibrated under drift, and generalizes to biological return, the present protocol should accept it. The requirement is a discriminating test, not interpretability for its own sake.

The strongest system-identification alternative argues that preparation-specific state-space estimation already provides the required intervention map. If that framework includes all seven operations and evaluates them as one protocol, the novelty claim should be narrowed to implementation, genealogy, or a particular rescue criterion.

The strongest compensatory alternative says that functional recovery is what matters, even when the original relation is not restored. In engineering or rehabilitation that may be correct. The paper distinguishes **functional compensation** from **mechanistic rescue** because they answer different questions. Compensation can be a success while failing to validate the nominated biological relation.

Retire or narrow the general claim if capacity-matched alternatives repeatedly equal the typed model on unseen lesions and wrong-rescue discrimination; typed maps cannot be estimated reproducibly; mappings fail under ordinary drift; signatures do not separate plausible mechanisms; or biological return experiments fail to distinguish the proposed relation from alternatives. Retire a local map when its matching-scale return fails, even if another part of the architecture remains useful.

12 Limitations, Safety, and Evidence Boundaries

The present application is deterministic and synthetic. Its variables are fully measured, its true mechanism is known, and the generator favors the correct typed model. Real biological systems contain hidden state, nonstationarity, stochasticity, multiscale feedback, measurement error, and ethical constraints. The current model therefore functions as an executable specification and failure-control demonstration.

The formal theorems are similarly bounded. They show why declared assumptions suffice for specific logical conclusions and why one observation cannot determine every intervention. They do not supply the assumptions. No theorem in this paper proves consciousness, sentience, biological identity, treatment efficacy, or intervention safety.

The proposed return tests must use the least invasive adequate preparation, external safety review, preregistered stopping rules, and explicit monitoring of off-target effects. Human applications would require a separate translational and regulatory program. The protocol’s purpose is to make hidden inference steps visible before an intervention, not to bypass them.

13 Discussion

Reciprocal translation reframes a common debate. Biological fidelity and engineering usefulness are not opposites. An abstraction can be useful while preserving only one relation, and that limitation can be scientifically productive when it is declared. The problem begins when endpoint success is allowed to imply a mechanism that has not survived an intervention.

The method also changes what counts as rescue. In a complex adaptive system, behavior can recover through multiple routes. A selective mechanistic rescue therefore needs a signature rich enough to reject a mean-matched alternative, plus off-target measurements that constrain spillover. The pilot operationalizes this point in the simplest setting: the wrong rescue recovers the mean but not the conditional relation.

SAN's distinctive contribution is the joined loop. Its early sources asked how one embodied network produces, senses, learns, and acts through its own changing patterns. The modern method converts that philosophical and architectural question into a scientific workflow. A biological mechanism is not merely copied into AI; it is exposed to controlled failures that may be difficult or unsafe to attempt first in biology. The engineered failure then earns scientific value only when it produces a discriminating, matching-scale return experiment.

This workflow is intentionally symmetric in revision. A failed biological return can mean that the engineered abstraction omitted a relevant state, that the map targeted the wrong scale, that the biological measurement was insufficient, or that the nominated mechanism was incorrect for the preparation. Both models change. The method does not protect the initial SAN interpretation from evidence; it protects the full source-faithful argument from strawman compression while making each scientific claim falsifiable at its proper scale.

14 Conclusion

AI-neuroscience translation becomes scientifically stronger when the full round trip is governed. Type the biological relation. Declare the partial map. Compare strong capacity-matched alternatives. Hold out perturbations. Distinguish correct rescue from relation-wrong endpoint recovery. Preserve off-target measurements. Return only the discriminating signature to a matching-scale biological experiment. Revise both systems.

The formal kernel and reference application show that these requirements can be expressed, executed, falsified, and audited. In the frozen synthetic family, the correct typed model generalized to unseen interventions, the wrong map failed, and a mean-matched rescue was exposed by conditional structure. Those are application results, not neuroscience findings. The paper's central claim will earn biological support only when one of the proposed return experiments completes the loop.

Historical Development, Authorship, and AI Assistance

I developed the Self-Aware Networks program and the precursor source materials cited here. The dated genealogy distinguishes owner dates, immutable public fixation, later synthesis, and external scientific evidence. The complete reciprocal protocol is dated to 2026 rather than projected backward into earlier notes.

AI systems assisted under my direction with source routing, literature discovery, formalization, code implementation, validation, editing, and document production. Their outputs were treated as work products requiring source checks and deterministic validation, not as independent scientific evidence. I am responsible for the claims and interpretation.

Data, Code, and Formal Materials

The release companion contains the frozen configuration, reference application, complete reported and rerun tables, generated figures, manifests, validators, validation receipts, Lean source, claim signature, compatibility ledger, compiler receipt, source ledger, and claim-source matrix. The application uses synthetic data and requires no personal data.

References

1. McCulloch WS, Pitts W. A logical calculus of the ideas immanent in nervous activity. *Bulletin of Mathematical Biophysics*. 1943;5:115-133. <https://doi.org/10.1007/BF02478259>
2. Rosenblatt F. The perceptron: a probabilistic model for information storage and organization in the brain. *Psychological Review*. 1958;65:386-408. <https://doi.org/10.1037/h0042519>
3. London M, Hausser M. Dendritic computation. *Annual Review of Neuroscience*. 2005;28:503-532. <https://doi.org/10.1146/annurev.neuro.28.061604.135703>
4. Branco T, Clark BA, Hausser M. Dendritic discrimination of temporal input sequences in cortical neurons. *Science*. 2010;329:1671-1675. <https://doi.org/10.1126/science.1189664>
5. Guerguiev J, Lillicrap TP, Richards BA. Towards deep learning with segregated dendrites. *eLife*. 2017;6:e22901. <https://doi.org/10.7554/eLife.22901>
6. Ziv Y, Burns LD, Cocker ED, et al. Long-term dynamics of CA1 hippocampal place codes. *Nature Neuroscience*. 2013;16:264-266. <https://doi.org/10.1038/nn.3329>
7. Driscoll LN, Pettit NL, Minderer M, Chettih SN, Harvey CD. Dynamic reorganization of neuronal activity patterns in parietal cortex. *Cell*. 2017;170:986-999.e16. <https://doi.org/10.1016/j.cell.2017.07.021>
8. Walker EY, Sinz FH, Cobos E, et al. Inception loops discover what excites neurons most using deep predictive models. *Nature Neuroscience*. 2019;22:2060-2065. <https://doi.org/10.1038/s41593-019-0517-x>
9. Ding Z, Tran D, Ponder K, et al. Functional bipartite invariance in mouse primary visual cortex receptive fields. *Nature Neuroscience*. 2026;29:851-863. <https://doi.org/10.1038/s41593-026-02213-3>
10. Ding Z, Fahey PG, Papadopoulos S, et al. Functional connectomics reveals general wiring rule in mouse visual cortex. *Nature*. 2025;640:459-469. <https://doi.org/10.1038/s41586-025-08840-3>
11. Draelos A, Loring MD, Nikitchenko M, et al. A software platform for real-time and adaptive neuroscience experiments. *Nature Communications*. 2025;16:9909. <https://doi.org/10.1038/s41467-025-64856-3>
12. Orsborn AL, Moorman HG, Overduin SA, et al. Closed-loop decoder adaptation shapes neural plasticity for skillful neuroprosthetic control. *Neuron*. 2014. <https://doi.org/10.1016/j.neuron.2014.04.048>
13. Bach-y-Rita P, Collins CC, Saunders FA, White B, Scadden L. Vision substitution by tactile image projection. *Nature*. 1969;221:963-964. <https://doi.org/10.1038/221963a0>
14. Ptito M, Moesgaard SM, Gjedde A, Kupers R. Cross-modal plasticity revealed by electrotactile stimulation of the tongue in the congenitally blind. *Brain*. 2005;128:606-614. <https://doi.org/10.1093/brain/awh380>
15. Wilson JA, Walton LM, Tyler M, Williams J. Lingual electrotactile stimulation as an alternative sensory feedback pathway for brain-computer interface applications. *Journal of Neural Engineering*. 2012;9:045007. <https://doi.org/10.1088/1741-2560/9/4/045007>
16. Beaubois R, Cheslet J, Duenki T, et al. BioemuS: A new tool for neurological disorders studies through real-time emulation and hybridization using biomimetic spiking neural network. *Nature Communications*. 2024;15:5142. <https://doi.org/10.1038/s41467-024-48905-x>
17. Toker D, et al. Adversarial AI reveals mechanisms and treatments for disorders of consciousness. *Nature*

- Neuroscience*. 2026;29:964-977. <https://doi.org/10.1038/s41593-026-02220-4>
18. Templeton A, et al. Scaling monosemanticity: extracting interpretable features from Claude 3 Sonnet. Anthropic Research. 2024. <https://www.anthropic.com/research/mapping-mind-language-model>
 19. Blumberg M. *Building Sentient Beings*. 2025. <https://doi.org/10.5281/zenodo.15522356>

SAN Source Genealogy

The following sources establish genealogy rather than independent scientific validation:

- Blumberg M. a0051z.md, owner-dated September 21, 2011; public Git fixation June 8, 2022. Sensation-prediction-action-return loop and “a learning loop that learns itself.”
- Blumberg M. a0632z.md, owner-dated April 19, 2012; public Git fixation June 8, 2022. Same-network production, reception, and learning.
- Blumberg M. *Neural Lace Podcast 1*, 2017; later Git-fixed transcript. Sending, receiving, connection, feedback, and Alternating Transmission Protocol.
- Blumberg M. b0327ywhisper.md, recording dated March 25, 2021; transcript Git-fixed January 23, 2024. Cortical traffic, decoding, and compatible sensory return question.
- Blumberg M. SA0v9.md, public Git fixation June 8, 2022. Neo Mind Cycle/BCI, biological SAN, and artificial-neural program map.
- Blumberg M. *Bridging Molecular Mechanisms and Neural Oscillatory Dynamics*, public launch October 30, 2024. Biological mechanism staircase.
- Blumberg M. *Building Sentient Beings*, 2025. Engineered embodiment, development, and self-modeling route.
- SAN Book 2 Cycle 37, frozen August 24, 2026. Mature reciprocal mechanism-transfer method.